Finding Value in MLB Pitchers

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Finding Value in Major League Baseball Pitchers: Who's Overpaid and Underpaid?

Audience

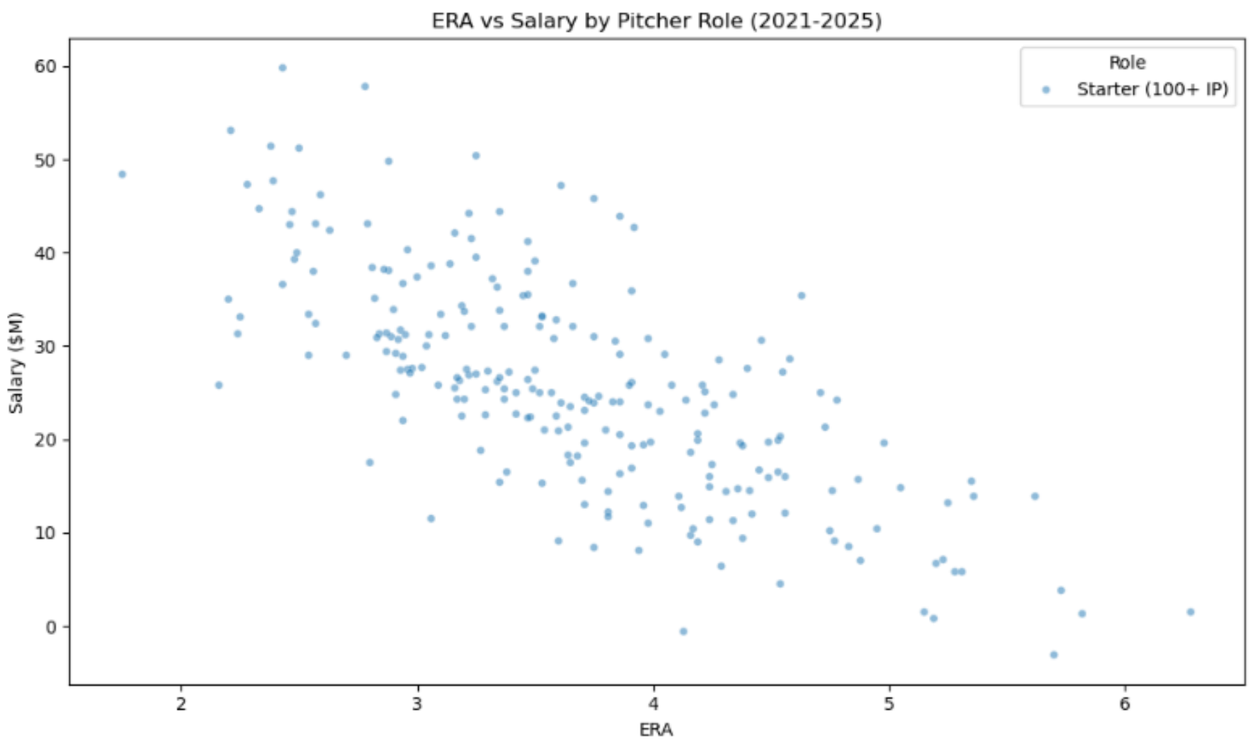
- MLB Front Office Analysts and General Managers evaluating Pitcher contracts and **targeting free agents or potential trades**

Motivation

- To investigate the relationship between pitching performance metrics and salary across the 2021-2025 seasons to **identify underpaid pitchers with strong potential**

Key Questions

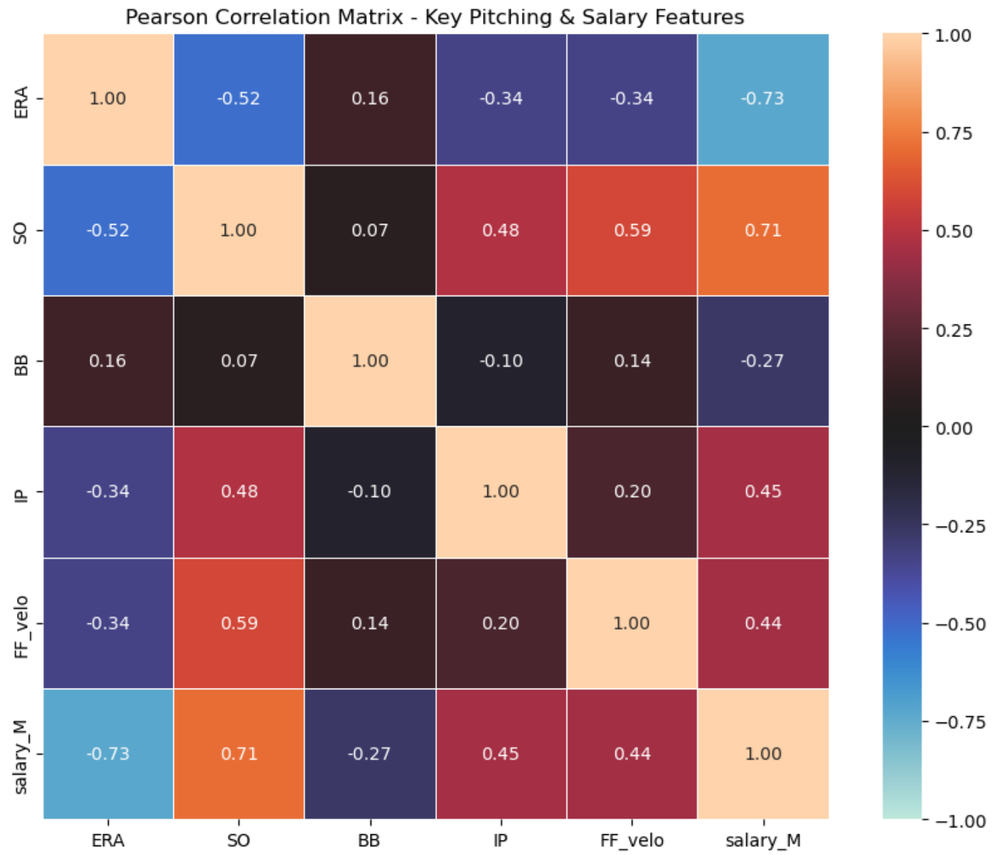
- Which metrics best predict ERA?
- Which metrics best predict salary?
- Which 2025 underpaid pitchers represent the best value this offseason?



238
PITCHER-SEASONS
2021–2025

3
DATA SOURCES
MLB API · Savant · FanGraphs

Pearson Correlation confirms that higher salaries are rewarded to those with lower ERAs, indicating a generally efficient trend across baseball.



	velo_tier	n_pitchers	avg_ERA	avg_K_BB	avg_salary_M
0	96+ mph (Elite)	41	3.33	4.34	33.89
1	93-95 mph (Above Avg)	114	3.60	3.64	25.06
2	90-92 mph (Average)	65	3.94	3.51	22.31
3	Below 90 mph	11	4.16	3.06	15.79

→ ***A lower ERA is correlated with a higher salary at -0.73 and with strikeouts at -0.52***
→ ***Those with higher velocities tend to have a stronger ERA, K/BB, and higher salary.***

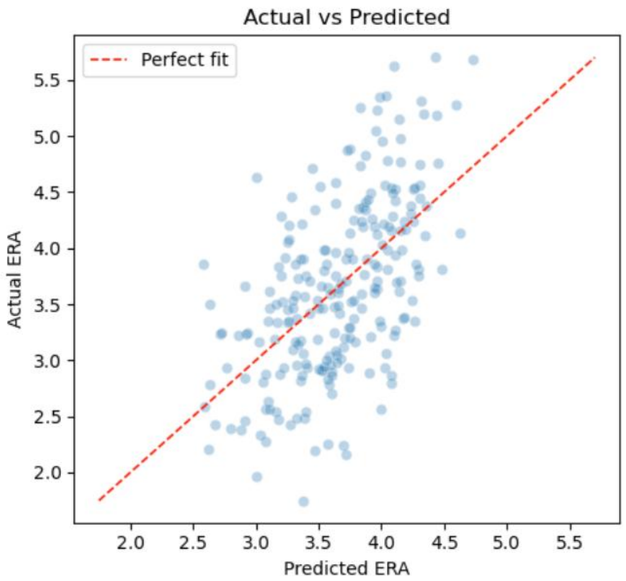
Two Models: Predicting ERA and Predicting Value

Model 1: Predicting ERA

R^2 = 0.33 | RMSE = 0.64

Features: SO, BB, FF velocity, K/BB ratio, GB%

ERA is noisy (defense, luck) — but strikeouts are the strongest predictor



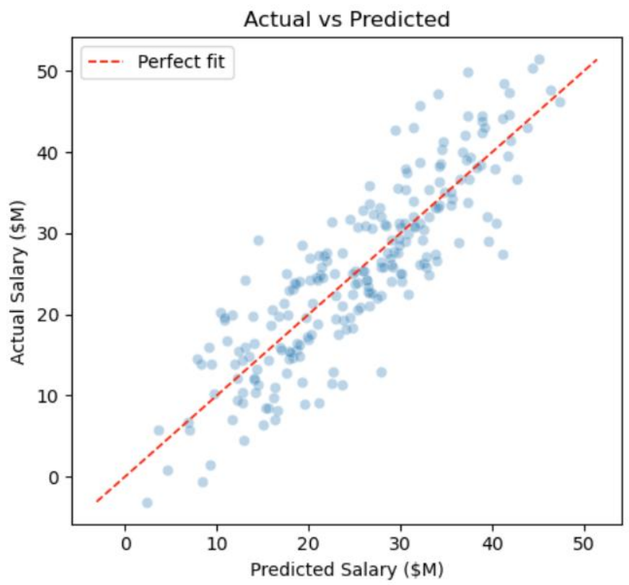
	Units
R^2	0.33
RMSE	0.64
Intercept	3.65
K Coef.	-0.41
BB Coef.	0.17
FF Velo Coef.	-0.03
K/BB Coef.	0.01
GB % Coef,	-0.10

Model 2: Predicting Value

R^2 = 0.75 | RMSE = $5.48M

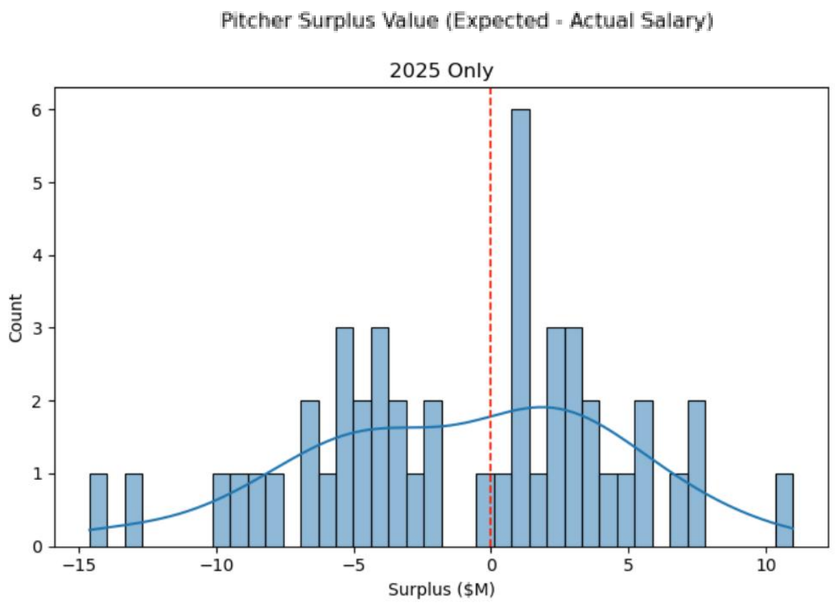
Features: ERA, SO, BB, FF velocity, K/BB ratio, GB%

SO and ERA are the dominant drivers of production value



	Units
R^2	0.75
RMSE	5.48
Intercept	25.3
ERA	-4.23
K Coef.	5.36
BB Coef.	-2.00
FF Velo Coef.	0.15
K/BB Coef.	1.24
GB % Coef,	1.92

The models predicted value for pitchers minus their actual salary showed that most pitchers cluster near zero, but the tails are where teams may find value.



TOP UNDERVALUED

Gavin Williams (CLE)	167.2 IP · 3.06 ERA · 173 SO	+$11.0M
Zack Littell (CIN)	186.2 IP · 3.81 ERA · 130 SO	+$7.5M
Bryan Woo (SEA)	186.2 IP · 2.94 ERA · 198 SO	+$7.5M

TOP OVERVALUED

Michael Wacha (KC)	172.2 IP · 3.86 ERA · 126 SO	−$14.6M
Jesús Luzardo (PHI)	183.2 IP · 3.92 ERA · 216 SO	−$13.2M
Luis Severino (OAK)	162.2 IP · 4.54 ERA · 124 SO	−$9.9M

Future Considerations:

- → *Incorporate Fielding-Independent Pitcher (FIP) or xERA as less-noisy measures than ERA*
- → *Leverage projection systems to identify top characteristics of pitchers who were under or overvalued*
- → *Use cluster analysis or PCA on pitchers and pitch types to control for difference in pitch deployment versus pitch quality*